

ESMI Applied Mathematics Project Syllabus

Queueing Theory

9 June – 6 August 2025

Instructor: Laurel Newman (lan66@cornell.edu)

Lecture Location: 146 Upson

Lecture Time:

- Weeks 1 & 2 (9 June - 20 June): 9:00 am - 12:00 pm M\T\W\R\F
- Weeks 3-9 (23 June - 6 August): 1:30-3:30 pm M\W\F

Office Hours: By appointment

1 Introduction

This summer we will gain experience in doing mathematical research and cover a brief introduction to queueing theory. We will cover probability, some relevant statistical distributions, and Python. Students will work in small research groups to develop their own question about a real-world queueing system and use mathematical and computational tools to gain insight into the system. Students will write up their results and present them at the end of the program.

1.1 Resources

Here are some additional resources you may wish to make use of if you wish to explore a topic in more depth than we cover in lecture.

Some further reading on Queueing Theory: https://www.cs.ucf.edu/~lboloni/Teaching/EEL6785_Fall2010/slides/QueueingTheory.pdf

NumPy for beginners: https://numpy.org/doc/stable/user/absolute_beginners.html

Introductory probability: <https://math.dartmouth.edu/~prob/prob/prob.pdf>

A great resource for brainstorming queueing problems: <https://opexlearning.com/resources/queueing-theory/>

1.2 Timeline

1. Monday 9 June: program starts, beginning first two weeks of daily lectures

2. Monday 23 June: students' summer courses begin, we now meet M/W/F.
 - Mondays: research skills and best practices
 - Wednesdays: work days
 - Fridays: work days
3. Friday 27 June: students finalize their research groups and submit their project proposals and group contracts
4. Thursday 24 July: students submit their draft research report and presentation. Submit a "working draft" of poster.
5. Thursday 31 July: "working draft" of final research report and presentation due. Final draft of poster: poster will be printed out in Mann library
6. Monday 4 August: students submit their final research report and presentation to ESMI Box Folder by start of class
7. Wednesday 6 August: Summer Research Symposium

1.3 Deliverable Expectations

All deliverables should be

- Turned in to the ESMI Box folders (not attempted to be adjusted off a USB last minute)
- Late submissions might not be up to date
- Titled appropriately and consistently: "Draft_TeamName_Poster.pdf" or "Final_TeamName_Slides.pptx"

1.4 Schedule

Week	Day	Lecture
1	9 June	Introduction
	10 June	Intro to Queues
	11 June	Non-determinism
	12 June	Combinatorics
	13 June	Intro to Probability
2	16 June	Probability II: Conditional Probability
	17 June	Probability III: Applying Probability
	18 June	Statistics
	19 June	Juneteenth No class
	20 June	Hawkes Processes
3	23 June	Intro to Research & Python
	25 June	Graphing your results & NumPy for Queueing Theory
	27 June	Project Proposals & Group Contracts Due & Presenting Your Results
4	30 June	Coding Documentation
	2 July	Project Planning
	3 July	Work Day
5	7 July	How to write up your results
	9 July	
	11 July	Work Day
6	14 July	Effective Math Communication
	16 July	
	18 July	Work Day
7	21 July	
	23 July	
	24 July	“Working draft” of poster, drafts of presentation and report due
	25 July	
8	28 July	
	30 July	
	31 July	Final draft of poster, “working draft” of presentation, report due
	1 August	Posters Printed, students give rehearsal talks
9	4 August	Final presentation slides and report due by start of class
	6 August	Symposium